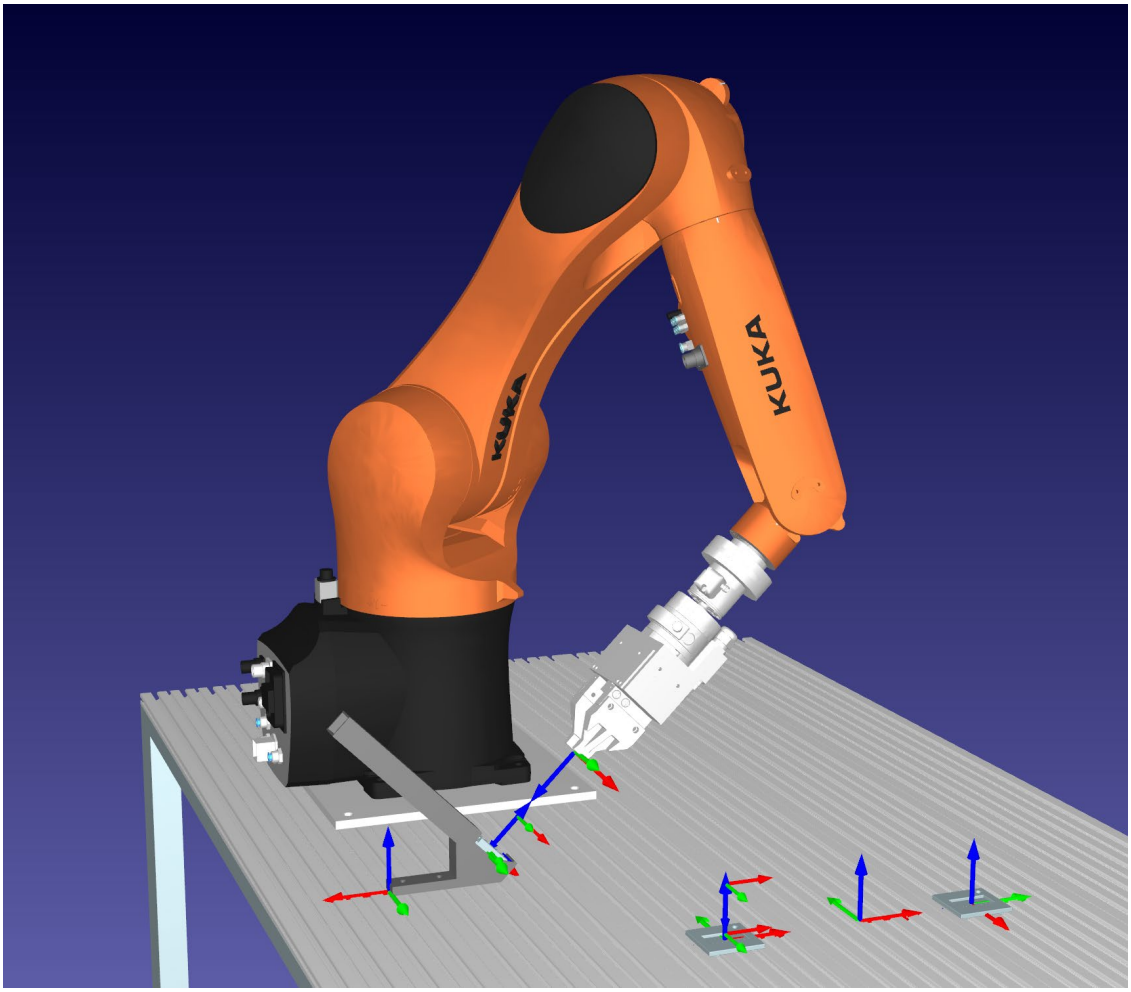
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Simulation of a KUKA Industrial Robot

Instruction Manual for Student Lab Exercise



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1. Introduction

Designing and testing a robotic cell can be a time consuming and costly process. This can make it hard for beginners to get started in the field of robotics, and it can slow down the development process for an experienced roboticist. Therefore, it is important to work with a simulation software, so you can thoroughly test your designs before any real resources are committed.

The purpose of this lab is to learn

a) how to simulate basic operations of an industrial robot in a robotics simulation software (RoboDK), and
b) how to transfer the simulation into the real world by running it on a KUKA industrial robot.

The experiment has two phases:

1) Preparation at home using RoboDK (generally, you will have two weeks)
2) Lab experiment with the real KUKA (90 minutes)

Some inspiration:

- How NASA uses RoboDK for multi-robot inspection: https://robodk.com/blog/multi-robot-inspection/
- How our robotics lab used RoboDK for a recycling application: https://robodk.com/blog/students-win-competition-with-robodk/

You will need some basic background about industrial robots and know the terms. If you need to brush up your background knowledge, refer to the KUKA material provided in **Appendix 1: Required Files**.

Watch the following video which shows the end result of what you need to achieve in your preparation before coming to the lab:

`simulation_KUKA.mkv`

2. Preparing RoboDK

Please watch this video:

- https://youtu.be/xZ2_JEbS_E0

Follow these steps:

1) Download and install RoboDK
2) Ensure you have all files listed in **Appendix 1: Required Files**

3. Setting up the Workspace

Please watch these videos:

- https://youtu.be/uxlfxgIE2YE
- https://youtu.be/gNgCwwpZrWE

Now follow these steps:

1) Create a new station and name it `kuka_station`.
2) Place the table
    a) Make a new frame and name it `table_frame`.
    b) Import the object `table_object.sld` into your station.
    c) Make the `table_frame` the reference frame of the `table_object`. Put the `table_frame` in the corner of the `table_object` by specifying an object position of `[0 0 0 0 0 0]`.

Make sure your workspace looks like this:

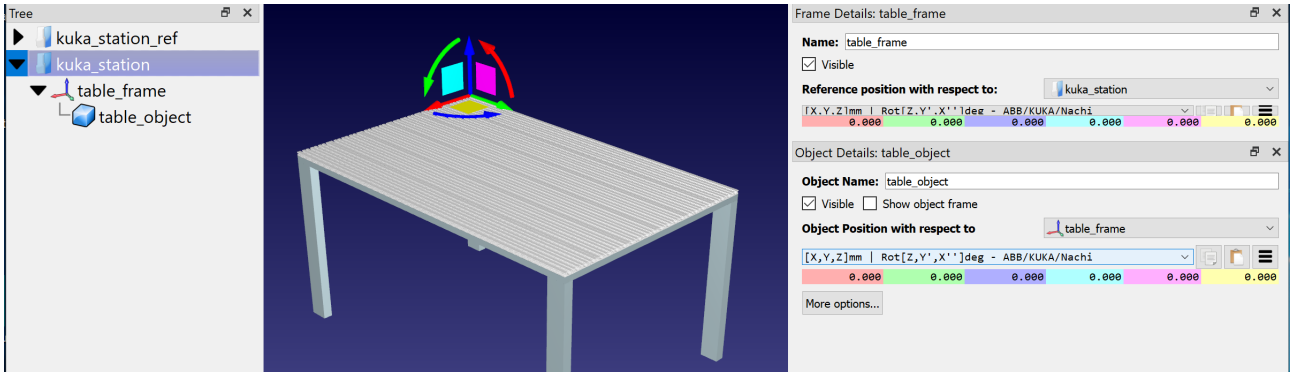


Figure 1: Workspace after importing the table_frame and table_object

3) Place all other objects
    a) Repeat step 2 from above to place all frames and objects in accordance with **Appendix 2: Positions of Frames and Objects**.
4) Place the KUKA robot
    a) Download the following robot from the online library:
    KUKA KR 6 R900 sixx, type: 6 DOF, payload: 6 kg
    b) Change the KUKA's reference frame to the `kuka_frame`
    c) Gripper
        - Import the `gripper.robot` file.
        - Attach the gripper to the KUKA (right-click and move nested)
        - Delete the `gripper Base` frame
    d) TCP
        - Right-click on KUKA and select "Add tool (TCP)"
        - Rename to `gripper_TCP`
        - Specify the TCP with respect to the KUKA (robot flange):
        `[0.0 0.0 288.0 43.0 0.0 0.0]`

Your station tree and workspace should now look like this:

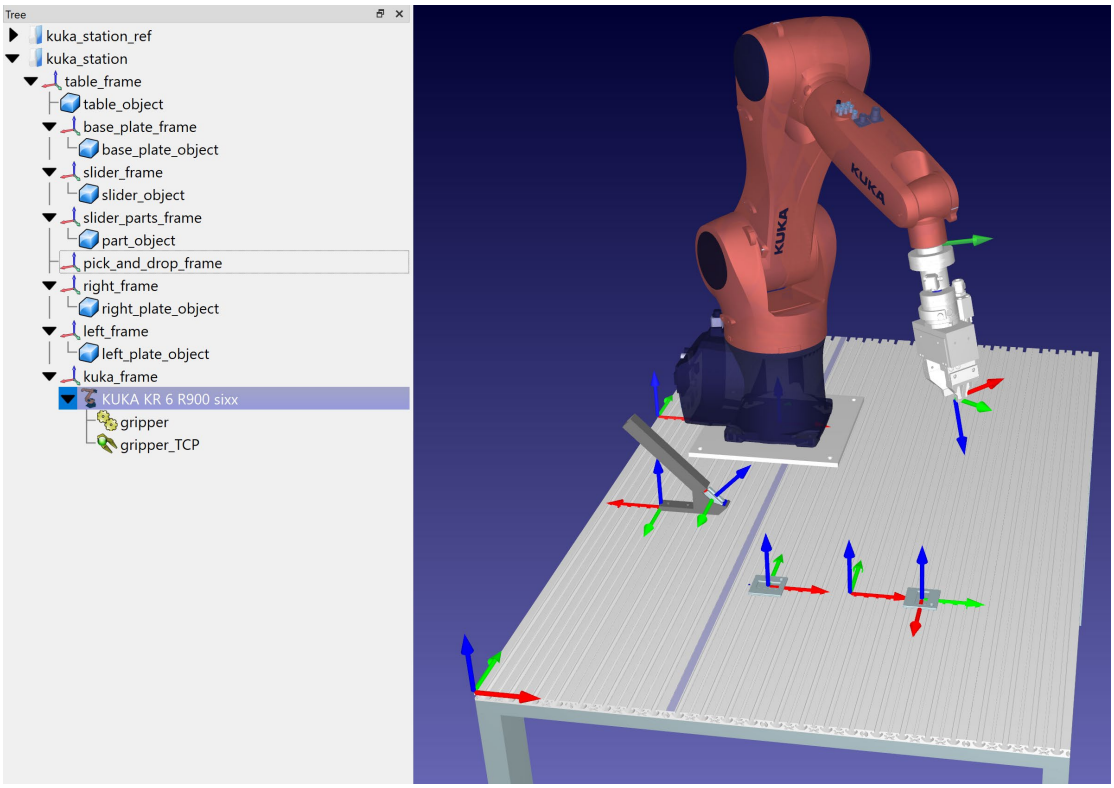


Figure 2: Station tree and workspace after setting up all frames and objects.

4. Creating a Recovery Program

Next, we will create our first robot program. This program allows resetting the workspace to its original settings, i.e. move objects and frames back to their original positions.

Please follow these steps:

1) Right-click on the station bar and choose `Add folder`.
2) Name the folder `programs`.
3) Select `Add program` (e.g. by using the symbol shown on the right)
4) Name the program `factory_reset`
5) Right-click on `factory_reset`, then click on:
   `Add instruction -> Simulation Event Instruction`.
6) From the drop-down menu, choose: `Set object position (relative)`.
7) Select all items in the dropdown-menu (CTRL + a) and press OK.



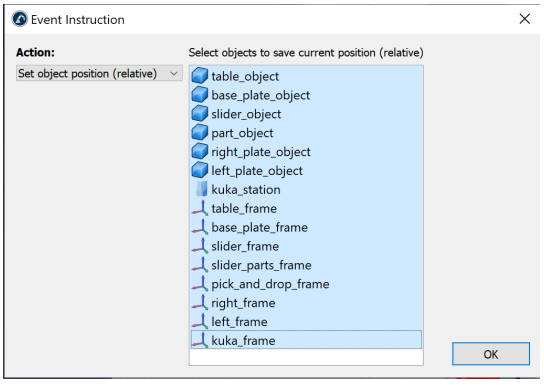


Figure 5: Creating a program for the factory reset.

5. Creating Targets

The next step is to create so-called targets. Targets are poses (position and orientation) to reach with the tool of the robot.

Please watch these videos to learn about targets:

- https://youtu.be/ilmJSD-a9bs
- https://youtu.be/ylKTq03-b_A

Let us create our first target, the `home_target`, which acts as a start and end point of our program:

1) Move the robot's TCP to your desired position, e.g. via teleoperation (`Joint axis jog`) or via dragging the TCP. You can also try the `Align` button to help align the robot with the table.

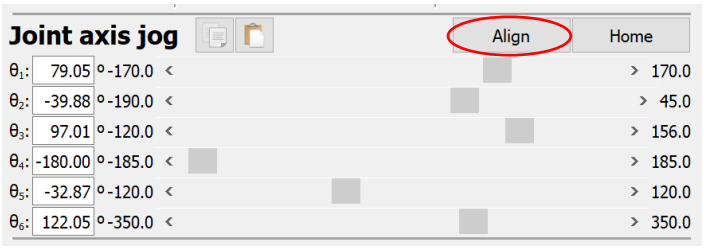


Figure 3: Teleoperation of the KUKA by commanding each axis (joint). Try the Align button!

2) Add a target and rename it to `home_target`.
3) Make sure the `home_target` is referenced to the `table_frame`, i.e. it should show up underneath the `table_frame` within the station tree on the left of your screen. If that is not the case, then right-click on target, select `Change support` and select `table_frame`.

Next, let us create targets to manipulate the part that is located on the left plate. It is recommended to create two targets:

1) `left_plate_target`: the target for picking and placing the part from/onto the left plate
   a) You can make the KUKA's TCP go to the `left_plate_object` using the `Get with robot` tool:

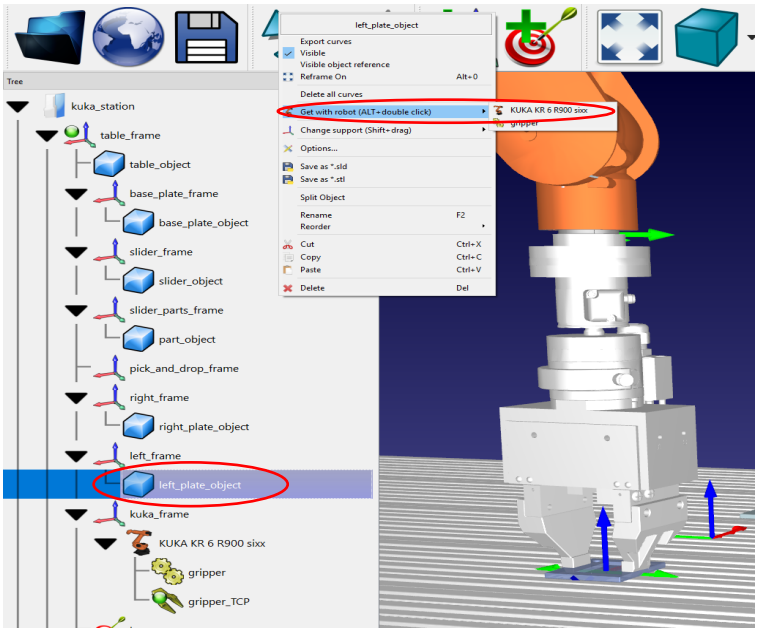


Figure 4: The "Get with robot" tool can be quite useful to define targets related to objects!

b) Create a target and name it `left_plate_target`. As for the `home_target`, ensure that the reference frame is the `table_frame`.

2) `app_left_plate_target`: "app" stands for approach. For pick-and-place applications, it is a good idea to define a position above the actual pick-and-place target for a defined approach before picking/placing
   a) Move the TCP straight up, e.g. by using the wheel tool shown below.

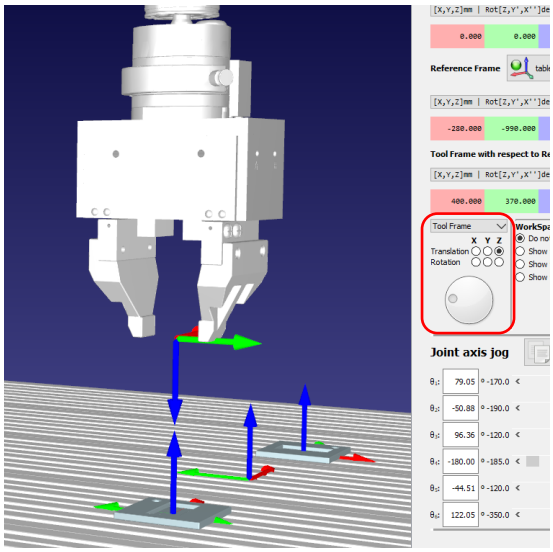


Figure 5: Using the wheel tool to move the TCP straight up

   b) Create a target in the `table_frame` and name it `app_left_plate_target`.

Now we have three targets. You can click on them and see the TCP of the robot moving between them. You can now imagine how those targets can be assembled into a robot program.

But before we get to that, you need to create a whole bunch of other targets as shown in **Appendix 3: Targets for the Application** (ignore the two green targets for now, only define the red targets).

Use the various tools that are available in the KUKA panel to define those targets. So far, we have learned the following methods:

- `Get with robot` tool
- `Align` tool
- Moving the TCP with the wheel in the tool frame

For the last method: it can be useful to change the reference frame to something other than the `table_frame`. In the example shown in Figure 6, you can see that the `slider_parts_frame` was chosen as a reference. This was used to define the approach target `app_slider_parts_target` that is 30mm in the z-direction of the angled `slider_parts_frame`.

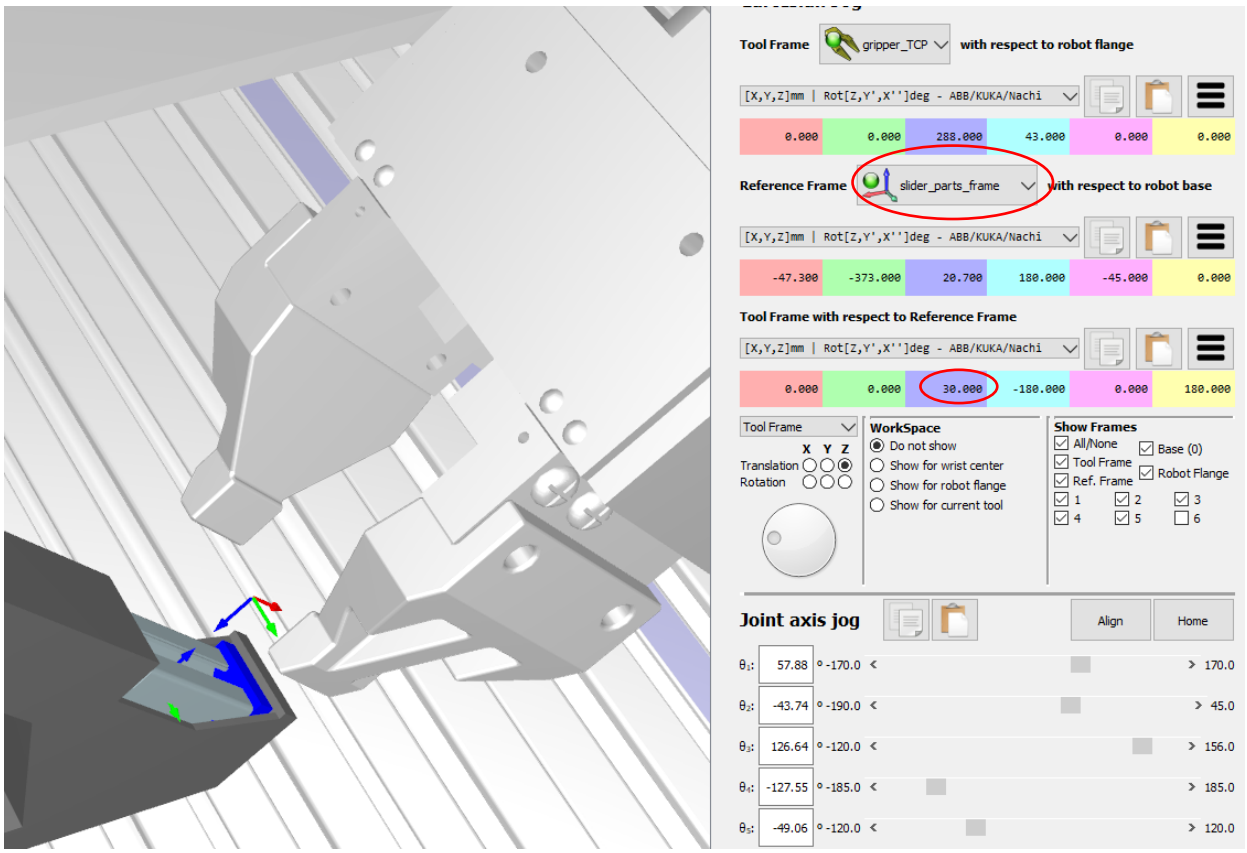


Figure 6: Make use of the previously defined frames to define targets. In this case, the angled slider_parts_frame was used to define the approach target app_slider_parts_target.

Note that for this case, the target will be created in the `slider_parts_frame` (check the station tree on the left-hand side of your screen). As already mentioned earlier, to move the target to the `table_frame`, select `Change support` and select `table_frame`. A faster solution is to change the reference frame back to the `table_frame`, then create the target.

Yet another tool that may come in useful is the `Other configurations` tool of the KUKA (see below). Roughly speaking, this setting determines what arrangements of joints the robot is allowed to take on. There may be more than one joint configuration for the robot to reach a 6D pose in space. In technical terms, the "inverse kinematic problem" can have multiple solutions. For this lab, simply play with the different configurations if necessary.

The figure below shows a summary of the tools you may want to use in order to define targets:

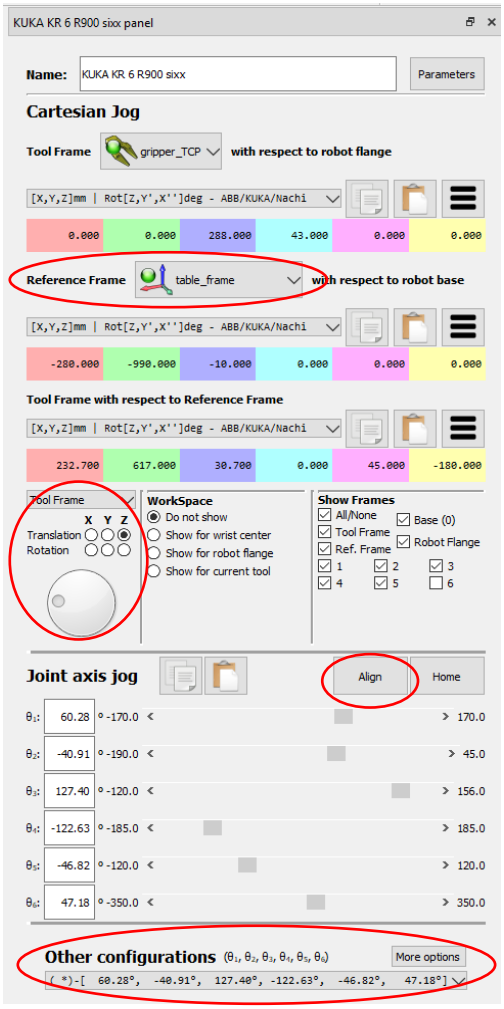


Figure 7: Useful tools to create targets

So far, we have defined "Cartesian targets" which are shown in red in in **Appendix 3: Targets for the Application**. For our application, we have to define two more targets which are used to open and close the gripper. Those are so-called "Joint targets" which are independent of the reference frame (read up on the RoboDK documentation if you want to know more).

Follow those steps to create the gripper targets:

1) `close_gripper_target`
    a. Double-click on the gripper symbol on the station tree. Enter d_1 = 20mm for the distance between the gripper fingers. This is the width of our `part_object`.

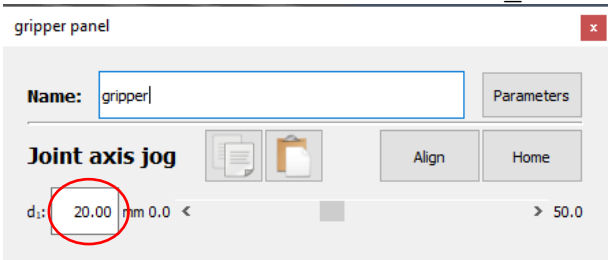


Figure 8: Gripper tool with a configuration for the distance between the fingers.

    b. Create a target, rename it to `close_gripper_target`, right-click on it, and define it as a Joint target.

c. Link the target to the `gripper` as shown below:

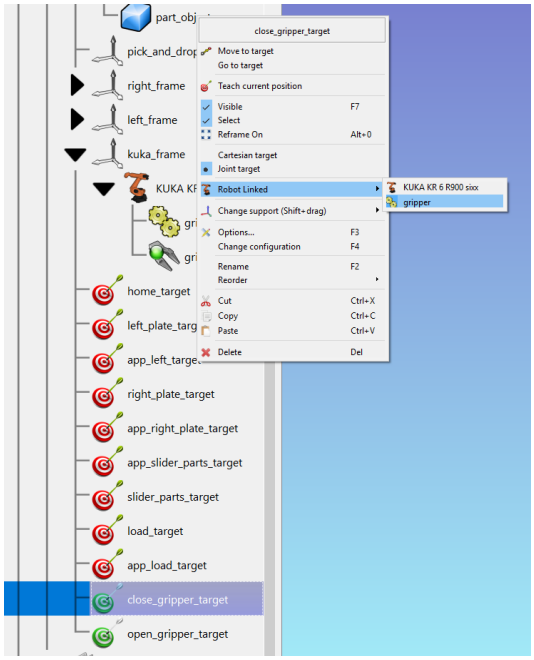


Figure 9: Linking the gripper to close_gripper_target

2) `open_gripper_target`: repeat the steps above with d_1=50mm
3) Create two programs to open and close the gripper
    a. As for the `factory_reset` program (see above), add the two programs to the already existing programs folder.
    b. Name them `open_gripper` and `close_gripper`.
    c. Right-click on the programs and add a `Move Joint Instruction` to the previously created targets (`open_gripper_target` and `close_gripper_target`).
    d. Link the programs to the `gripper` (same as Figure 9 but now for the program, not the target).

6. Creating the Robot Program

Please watch this video:

- https://youtu.be/38Yzhsom54E

Optionally, you can watch an entire webinar on a pick-and-place application:

- https://youtu.be/P0rEwpmAHq4

It is time to create the pick-and-place program using your previously created targets. Follow these steps:

1. Add a program called `main` to your programs folder (for this application, all "code" can be put into a single main program).
2. Watch the KUKA video again (`simulation_KUKA.mkv`) which shows the required motions of the robot program.
3. Put the program together. Some hints:
    a. You may have to adapt the coordinates of your targets (e.g. change Z to avoid crashing into the table), and/or introduce additional targets.
    b. Call the previously created subprograms `open_gripper` and `close_gripper` for opening and closing the gripper.
    c. To actually move the `part_object` in the simulation, you need to attach and detach it from the gripper (or more precisely the TCP) using the following function in your program:

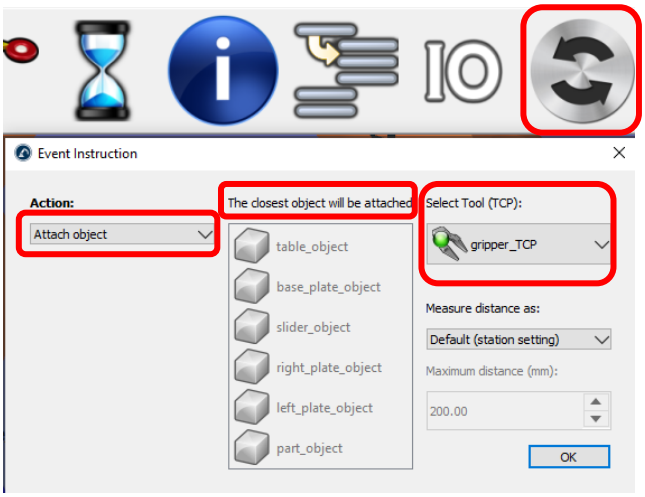


Figure 10: Use an Event Instruction to attach/detach the part_object from the gripper.

4. If you accidentally move the `part_object` or any other object and want them back in their original positions, you can use the `factory_reset` program.
5. Once done, generate the robot program as KUKA code and save the file as `main.src`:

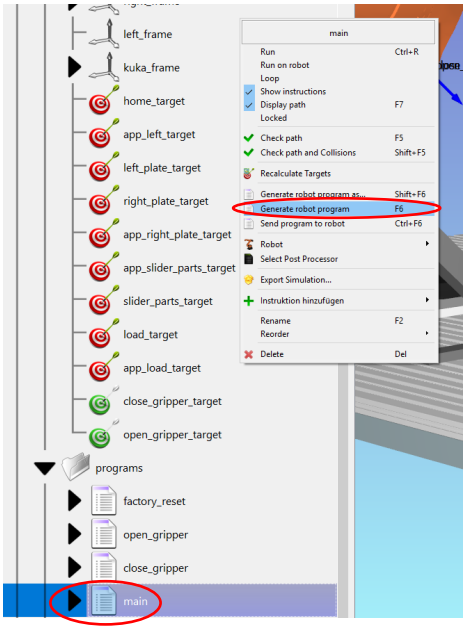


Figure 11: Creating the KUKA code

7. In the Lab

All participants need to bring the following to the lab session:

- Your laptop with RoboDK installed and a working simulation (you will fail the lab if the simulation is not working).
- On a USB stick:
  - RoboDK Station Module file (e.g. `kuka_station.rdk`)
  - KUKA program (e.g. `main.src`)

In the lab, the task is to port the application from the simulation world to the real robot. You will receive an introduction into safely running the KUKA, then run your program using the KUKA teach pendant.

As a preparation, it is a good idea to become familiar with the operation of the KUKA by browsing the KUKA material provided (see **Appendix 1: Required Files**).

Appendix 1: Required Files

The following files are available on E-learning:

File name	**Description**
`simulation_IR_KUKA_v2.0.pdf`	This manual
`EN_R_Versuchsbeschreibung, Robotik_KUKA.pdf`	Instruction manual for operating the KUKA industrial robot in the lab
`KSS_83_END_en.pdf`	KUKA operating manual
`simulation_KUKA.mkv`	Video showing the goal of the lab exercise
`gripper.robot`	RoboDK file
`table_object.sld`	RoboDK file
`base_plate_object.sld`	RoboDK file
`slider_object.sld`	RoboDK file
`left_plate_object.sld`	RoboDK file
`right-plate_object.sld`	RoboDK file
`part_object.sld`	RoboDK file

Appendix 2: Positions of Frames and Objects

Table 1: Positions of all frames and objects with reference to the `table_frame`.
***Important note:** select ABB/KUKA/Nachi as a format for the 6D pose as shown below!*

Frame /Object	**Translation in (mm):**			**Rotation around (deg):**		
	[X,Y,Z]mm | Rot[Z,Y',X'']deg - ABB/KUKA/Nachi					
	X – axis	**Y – axis**	**Z – axis**	**Z – axis**	**Y – axis**	**X – axis**
table_object	0.0	0.0	0.0	0.0	0.0	0.0
base_plate_frame	280.0	990.0	0.0	-90.0	0.0	0.0
base_plate_object	280.0	990.0	0.0	-90.0	0.0	0.0
slider_frame	125.0	617.0	0.0	180.0	0.0	0.0
slider_object	125.0	617.0	0.0	180.0	0.0	0.0
slider_parts_frame	232.7	617.0	30.7	180.0	-45.0	0.0
part_object	232.3	617.0	29.7	-180.0	-45.0	0.0
pick_and_drop_frame	550.0	370.0	0.0	0.0	0.0	0.0
right_frame	680.0	370.0	0.0	-90.0	0.0	0.0
right_plate_object	680.0	370.0	0.0	-90.0	0.0	0.0
left_frame	400.0	370.0	0.0	0.0	0.0	0.0
left_plate_object	400.0	370.0	0.0	0.0	0.0	0.0
kuka_frame	280.0	990.0	10.0	0.0	0.0	0.0

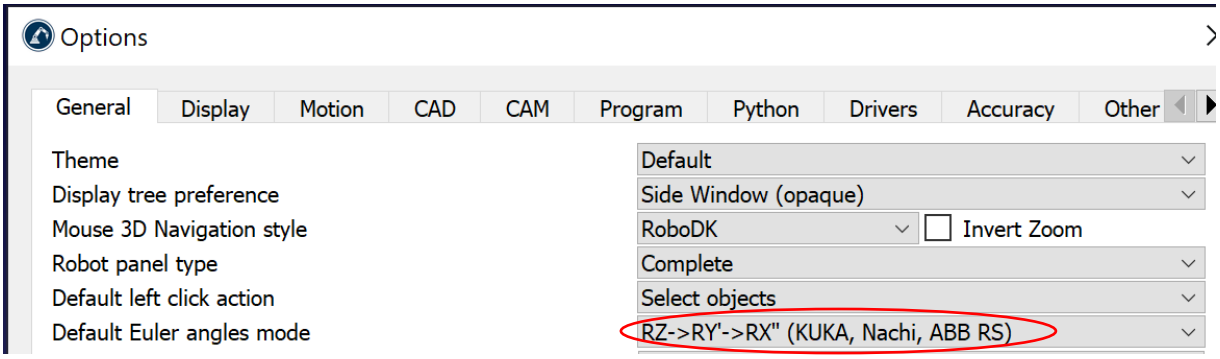


*Figure 12: **Hint**: in the Tools->Options menu, you can set the ABB/KUKA/Nachi format as a default.*

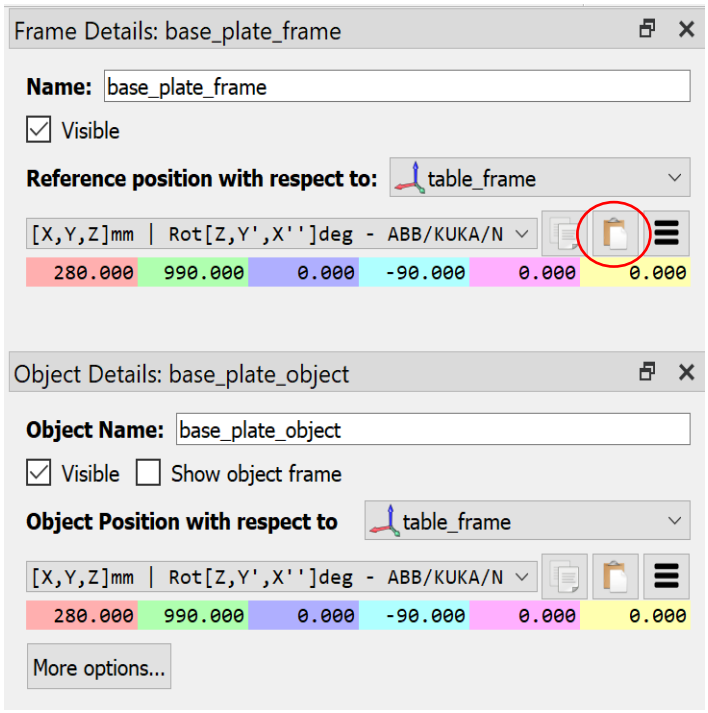


Figure 13: Example for the base plate.

***Hint**: Try to change the reference frame of the object (2nd window) to "base_plate_frame": what coordinates do you expect?*

***Hint 2**: You can directly paste the numbers from this document (see red circle above)*

Appendix 3: Targets for the Application

